Milestone4Extended.ipynb

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cind-820-learning-analytics

reports

sample_data

Disk 87.74 GB available

[1] 4s

```
from pathlib import Path
from time import perf_counter

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from scipy.stats import wilcoxon

from sklearn.compose import ColumnTransformer
from sklearn.ensemble import RandomForestClassifier
from sklearn.impute import SimpleImputer
from sklearn.inspection import permutation_importance
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    ConfusionMatrixDisplay,
    RocCurveDisplay,
    accuracy_score,
    f1_score,
    precision_score,
    recall_score,
    roc_auc_score
)
from sklearn.model_selection import (
    StratifiedKFold,
    cross_validate,
    train_test_split
)
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.tree import DecisionTreeClassifier
```

[2] 2s

```
!git clone https://github.com/britmap-codes/cind-820-learning-analytics.git
```

```
Cloning into 'cind-820-learning-analytics'...
remote: Enumerating objects: 97, done.
remote: Counting objects: 100% (97/97), done.
remote: Compressing objects: 100% (81/81), done.
remote: Total 97 (delta 28), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (97/97), 9.56 MiB | 13.09 MiB/s, done.
Resolving deltas: 100% (28/28), done.
```

[3] 0s

```
PROJECT_ROOT = Path.cwd()

if PROJECT_ROOT.name == "notebooks":
    PROJECT_ROOT = PROJECT_ROOT.parent

processed_path = PROJECT_ROOT / "data" / "processed"
fig_path = PROJECT_ROOT / "reports" / "figures"
table_path = PROJECT_ROOT / "reports" / "tables"

fig_path.mkdir(parents=True, exist_ok=True)
table_path.mkdir(parents=True, exist_ok=True)

print("Project root:", PROJECT_ROOT)
print("Processed data path:", processed_path)
print("Figures path:", fig_path)
print("Tables path:", table_path)
```

```
Project root: /content
Processed data path: /content/data/processed
Figures path: /content/reports/figures
Tables path: /content/reports/tables
```

[5] 0s

```
from pathlib import Path

PROJECT_ROOT = Path("/content/cind-820-learning-analytics")

processed_path = PROJECT_ROOT / "data" / "processed"

df = pd.read_csv(processed_path / "oulad_student_level.csv")
```

[6] 0s

```
df = pd.read_csv(processed_path / "oulad_student_level.csv")

if "at_risk" not in df.columns:
    df["at_risk"] = df["final_result"].map({
        "Fail": 1,
        "Withdrawn": 1,
        "Withdraw": 1,
        "Pass": 0,
        "Distinction": 0
    })

df = df.dropna(subset=["at_risk"]).copy()
df["at_risk"] = df["at_risk"].astype(int)

print("Dataset shape:", df.shape)
display(df.head())
```

```
Dataset shape: (32593, 16)
```

	code_module	code_presentation	id_student	gender	region	highest_education	imd_band	age_band	num_of_prev_attempts	studied_credits	c
0	AAA	2013J	11391	M	East Anglian Region	HE Qualification	90-100%	55<=	0	240	
1	AAA	2013J	28400	F	Scotland	HE Qualification	20-30%	35-55	0	60	
2	AAA	2013J	30268	F	North Western Region	A Level or Equivalent	30-40%	35-55	0	60	
3	AAA	2013J	31604	F	South East Region	A Level or Equivalent	50-60%	35-55	0	60	
					West						

4	AAA	2013J	32885	F	Midlands Region	Lower Than A Level	50-60%	0-35	0	60

[7] ✓ 0s

```
print("Duplicate rows:", df.duplicated().sum())
print("Missing values by column:")
display(df.isna().sum().sort_values(ascending=False).to_frame("missing_count"))

print("Target distribution:")
display(df["at_risk"].value_counts().rename_axis("at_risk").to_frame("count"))

```

```
Duplicate rows: 0
Missing values by column:
```

	missing_count
mean_score	6773
total_assessments	6750
mean_weight	6750
imd_band	1111
gender	0
code_module	0
id_student	0
code_presentation	0
age_band	0
highest_education	0
region	0
num_of_prev_attempts	0
final_result	0
disability	0
studied_credits	0
at_risk	0

```
Target distribution:
```

at_risk	count
1	17208
0	15385

[8] ✓ 0s

```
numeric_candidates = [
    "num_of_prev_attempts",
    "studied_credits",
    "mean_score",
    "total_assessments",
    "mean_weight"
]

numeric_cols = [column for column in numeric_candidates if column in df.columns]

numeric_df = df[numeric_cols].apply(pd.to_numeric, errors="coerce")

descriptive_summary = pd.DataFrame({
    "count": numeric_df.count(),
    "mean": numeric_df.mean(),
    "median": numeric_df.median(),
    "mode": numeric_df.mode().iloc[0],
    "std": numeric_df.std(),
    "variance": numeric_df.var(),
    "min": numeric_df.min(),
    "max": numeric_df.max()
}).round(4)

display(descriptive_summary)

descriptive_summary.to_csv(
    table_path / "final_descriptive_statistics.csv"
)
```

	count	mean	median	mode	std	variance	min	max
num_of_prev_attempts	32593	0.1632	0.0	0.0000	0.4798	0.2302	0.0	6.0
studied_credits	32593	79.7587	60.0	60.0000	41.0719	1686.9010	30.0	655.0
mean_score	25820	72.7683	76.0	80.0000	16.3750	268.1405	0.0	100.0
total_assessments	25843	6.7296	7.0	12.0000	3.7714	14.2233	1.0	14.0
mean_weight	25843	13.3508	9.9	8.3333	8.8234	77.8527	0.0	100.0

Next steps: New interactive sheet

[9] ✓ 0s

```
categorical_candidates = [
    "code_module",
    "code_presentation",
    "gender",
    "region",
    "highest_education",
    "imd_band",
    "age_band",
    "disability"
]

categorical_cols = [
    column for column in categorical_candidates
    if column in df.columns
]

X = df[numeric_cols + categorical_cols]
y = df["at_risk"]

numeric_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
```

```
    ("scaler", StandardScaler())
])

categorical_pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer([
    ("numeric", numeric_pipeline, numeric_cols),
    ("categorical", categorical_pipeline, categorical_cols)
])

print("Numeric features:", numeric_cols)
print("Categorical features:", categorical_cols)
```

```
Numeric features: ['num_of_prev_attempts', 'studied_credits', 'mean_score', 'total_assessments', 'mean_weight']
Categorical features: ['code_module', 'code_presentation', 'gender', 'region', 'highest_education', 'imd_band', 'age_band', 'disability']
```

[10] ✓ 0s

```
log_reg_pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("model", LogisticRegression(
        max_iter=2000,
        class_weight="balanced"
    ))
])

tree_pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("model", DecisionTreeClassifier(
        max_depth=5,
        class_weight="balanced",
        random_state=42
    ))
])

rf_pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("model", RandomForestClassifier(
        n_estimators=200,
        class_weight="balanced",
        random_state=42,
        n_jobs=-1
    ))
])

models = {
    "Logistic Regression": log_reg_pipeline,
    "Decision Tree": tree_pipeline,
    "Random Forest": rf_pipeline
}
```

[11] ✓ 0s

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    stratify=y,
    random_state=42
)

print("Training rows:", len(X_train))
print("Testing rows:", len(X_test))
```

```
Training rows: 26074
Testing rows: 6519
```

[12] ✓ 38s

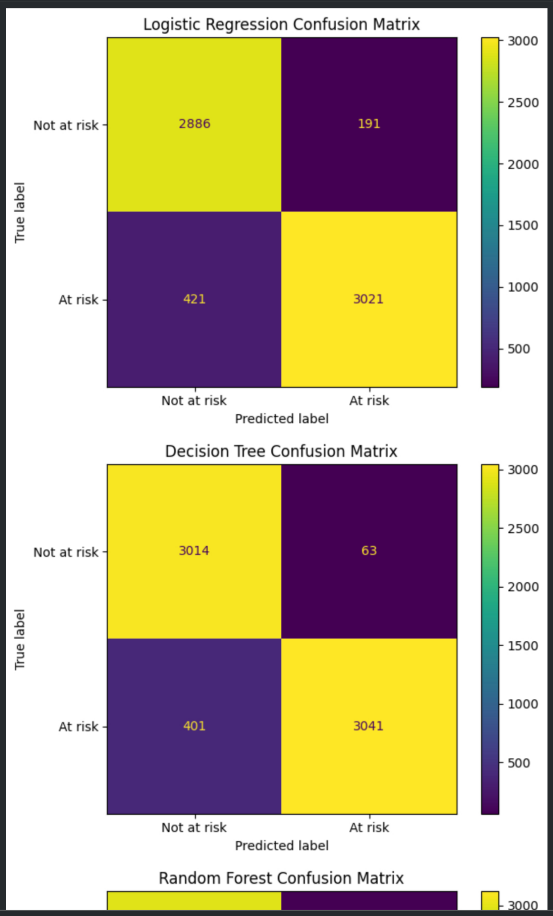
```
results = []
fitted_models = {}

for model_name, pipeline in models.items():
    start = perf_counter()
    pipeline.fit(X_train, y_train)
    train_time = perf_counter() - start

    start = perf_counter()
    y_pred = pipeline.predict(X_test)
    prediction_time = perf_counter() - start

    y_prob = pipeline.predict_proba(X_test)[:, 1]

    results.append({
        "Model": model_name,
        "Accuracy": accuracy_score(y_test, y_pred),
        "Precision": precision_score(
            y_test, y_pred, zero_division=0
        ),
        "Recall": recall_score(
            y_test, y_pred, zero_division=0
        ),
        "F1": f1_score(
            y_test, y_pred, zero_division=0
        ),
        "ROC_AUC": roc_auc_score(y_test, y_prob),
        "Train_time_seconds": train_time,
        "Prediction_time_seconds": prediction_time
    })

    fitted_models[model_name] = pipeline

test_results = pd.DataFrame(results).round(4)

display(test_results)

test_results.to_csv(
    table_path / "final_test_and_efficiency_results.csv",
    index=False
)
```

	Model	Accuracy	Precision	Recall	F1	ROC_AUC	Train_time_seconds	Prediction_time_seconds
0	Logistic Regression	0.9061	0.9405	0.8777	0.9080	0.9514	0.4593	0.0936
1	Decision Tree	0.9288	0.9797	0.8835	0.9291	0.9754	0.5638	0.1081
2	Random Forest	0.9356	0.9690	0.9070	0.9370	0.9786	36.5098	0.2436

Next steps: New interactive sheet

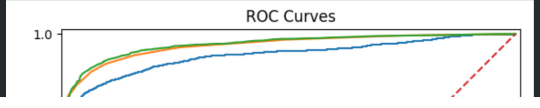
[13] ✓ 2s

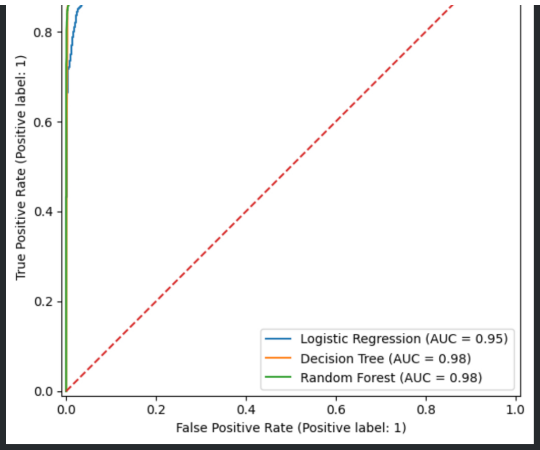
```python
for model_name, pipeline in fitted_models.items():
    y_pred = pipeline.predict(X_test)

    ConfusionMatrixDisplay.from_predictions(
        y_test,
        y_pred,
        display_labels=["Not at risk", "At risk"]
    )

    plt.title(f"{model_name} Confusion Matrix")
    plt.tight_layout()

    file_name = model_name.lower().replace(" ", "_")

    plt.savefig(
        fig_path / f"{file_name}_confusion_matrix.png",
        dpi=300,
        bbox_inches="tight"
    )

    plt.show()
```



[14] ✓ 1s

```python
plt.figure(figsize=(7, 6))

for model_name, pipeline in fitted_models.items():
    y_prob = pipeline.predict_proba(X_test)[:, 1]

    RocCurveDisplay.from_predictions(
        y_test,
        y_prob,
        name=model_name,
        ax=plt.gca()
    )

plt.plot([0, 1], [0, 1], linestyle="--")
plt.title("ROC Curves")
plt.tight_layout()

plt.savefig(
    fig_path / "combined_roc_curves.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```





[17] ✓ 3m

```python
cv = StratifiedKFold(
    n_splits=5,
    shuffle=True,
    random_state=42
)

scoring = {
    "accuracy": "accuracy",
    "precision": "precision",
    "recall": "recall",
    "f1": "f1",
    "roc_auc": "roc_auc"
}

cv_rows = []
fold_rows = []

for model_name, pipeline in models.items():
    scores = cross_validate(
        pipeline,
        X,
        y,
        cv=cv,
        scoring=scoring,
        n_jobs=-1
    )

    cv_rows.append({
        "Model": model_name,
        "Mean_Accuracy": scores["test_accuracy"].mean(),
        "SD_Accuracy": scores["test_accuracy"].std(),
        "Mean_Precision": scores["test_precision"].mean(),
        "SD_Precision": scores["test_precision"].std(),
        "Mean_Recall": scores["test_recall"].mean(),
        "SD_Recall": scores["test_recall"].std(),
        "Mean_F1": scores["test_f1"].mean(),
        "SD_F1": scores["test_f1"].std(),
        "Mean_ROC_AUC": scores["test_roc_auc"].mean(),
        "SD_ROC_AUC": scores["test_roc_auc"].std()
    })

    for fold_number in range(5):
        fold_rows.append({
            "Model": model_name,
            "Fold": fold_number + 1,
            "Accuracy": scores["test_accuracy"][fold_number],
            "Precision": scores["test_precision"][fold_number],
            "Recall": scores["test_recall"][fold_number],
            "F1": scores["test_f1"][fold_number],
            "ROC_AUC": scores["test_roc_auc"][fold_number]
        })

cv_results = pd.DataFrame(cv_rows).round(4)
fold_results = pd.DataFrame(fold_rows).round(4)

display(cv_results)
display(fold_results)

cv_results.to_csv(
    table_path / "final_cv_results.csv",
    index=False
)

fold_results.to_csv(
    table_path / "final_fold_scores.csv",
    index=False
)
```

	Model	Mean_Accuracy	SD_Accuracy	Mean_Precision	SD_Precision	Mean_Recall	SD_Recall	Mean_F1	SD_F1	Mean_ROC_AUC	SD_ROC_AUC
0	Logistic Regression	0.9093	0.0012	0.9402	0.0033	0.8845	0.0044	0.9115	0.0014	0.9530	0.0017
1	Decision Tree	0.9318	0.0038	0.9765	0.0025	0.8923	0.0079	0.9325	0.0040	0.9747	0.0006
2	Random Forest	0.9365	0.0042	0.9678	0.0038	0.9099	0.0050	0.9380	0.0041	0.9788	0.0007

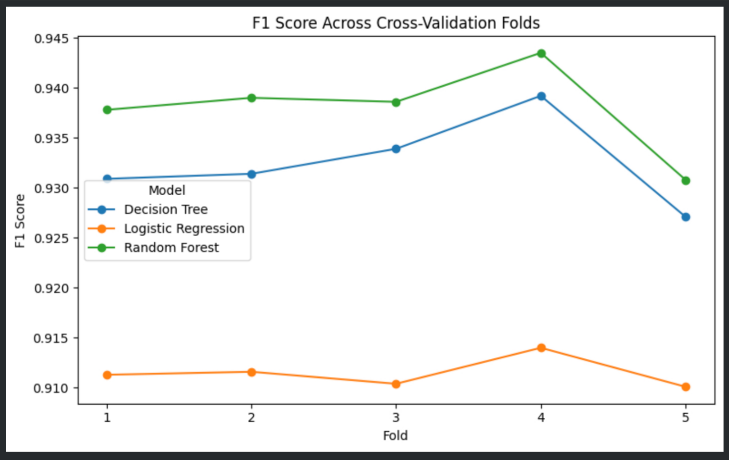
	Model	Fold	Accuracy	Precision	Recall	F1	ROC_AUC
0	Logistic Regression	1	0.9096	0.9459	0.8791	0.9113	0.9527
1	Logistic Regression	2	0.9095	0.9415	0.8835	0.9116	0.9511
2	Logistic Regression	3	0.9081	0.9378	0.8847	0.9104	0.9528
3	Logistic Regression	4	0.9113	0.9366	0.8925	0.9140	0.9561
4	Logistic Regression	5	0.9079	0.9391	0.8829	0.9101	0.9525
5	Decision Tree	1	0.9302	0.9749	0.8908	0.9309	0.9739

6	Decision Tree	2	0.9308	0.9770	0.8899	0.9314	0.9750
7	Decision Tree	3	0.9334	0.9811	0.8911	0.9339	0.9748
8	Decision Tree	4	0.9380	0.9738	0.9070	0.9392	0.9743
9	Decision Tree	5	0.9267	0.9759	0.8829	0.9271	0.9755
10	Random Forest	1	0.9365	0.9705	0.9073	0.9378	0.9784
11	Random Forest	2	0.9376	0.9694	0.9105	0.9390	0.9794
12	Random Forest	3	0.9371	0.9682	0.9108	0.9386	0.9779
13	Random Forest	4	0.9420	0.9705	0.9180	0.9435	0.9799
14	Random Forest	5	0.9291	0.9604	0.9029	0.9308	0.9786

Next steps: New interactive sheet New interactive sheet

[18] ✓ 0s

```python
fold_plot = fold_results.pivot(
    index="Fold",
    columns="Model",
    values="F1"
)

fold_plot.plot(
    marker="o",
    figsize=(8, 5)
)

plt.title("F1 Score Across Cross-Validation Folds")
plt.xlabel("Fold")
plt.ylabel("F1 Score")
plt.xticks(fold_plot.index)
plt.tight_layout()

plt.savefig(
    fig_path / "cv_f1_by_fold.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()

```



[19] ✓ 0s

```python
rf_f1 = (
    fold_results[
        fold_results["Model"] == "Random Forest"
    ]
    .sort_values("Fold")["F1"]
    .to_numpy()
)

log_f1 = (
    fold_results[
        fold_results["Model"] == "Logistic Regression"
    ]
    .sort_values("Fold")["F1"]
    .to_numpy()
)

tree_f1 = (
    fold_results[
        fold_results["Model"] == "Decision Tree"
    ]
    .sort_values("Fold")["F1"]
    .to_numpy()
)

rf_vs_log_stat, rf_vs_log_p = wilcoxon(rf_f1, log_f1)
rf_vs_tree_stat, rf_vs_tree_p = wilcoxon(rf_f1, tree_f1)

statistical_results = pd.DataFrame([
    {
        "Comparison": "Random Forest vs Logistic Regression",
        "Metric": "F1",
        "Statistic": rf_vs_log_stat,
        "P_value": rf_vs_log_p,
        "Mean_difference": (rf_f1 - log_f1).mean()
    },
    {
        "Comparison": "Random Forest vs Decision Tree",
        "Metric": "F1",
        "Statistic": rf_vs_tree_stat,
        "P_value": rf_vs_tree_p,
        "Mean_difference": (rf_f1 - tree_f1).mean()
    }
]).round(5)
```

```
44
45 display(statistical_results)
46
47 statistical_results.to_csv(
48     table_path / "final_statistical_comparisons.csv",
49     index=False
50 )
51
52
```

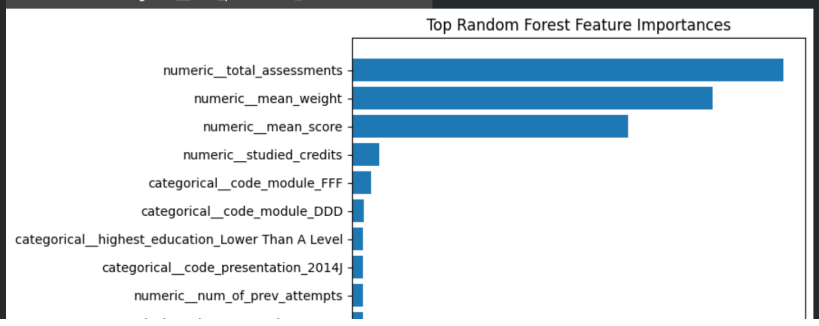
	Comparison	Metric	Statistic	P_value	Mean_difference
0	Random Forest vs Logistic Regression	F1	0.0	0.0625	0.02646
1	Random Forest vs Decision Tree	F1	0.0	0.0625	0.00544

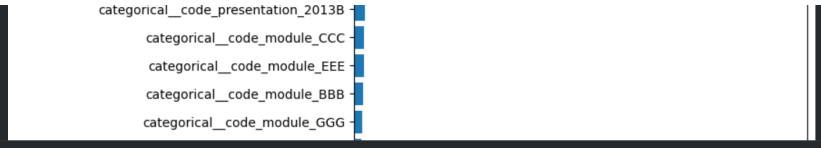
Next steps: New interactive sheet

[20] ✓ 2s

```
1 rf_fitted = fitted_models["Random Forest"]
2
3 feature_names = (
4     rf_fitted
5     .named_steps["preprocessor"]
6     .get_feature_names_out()
7 )
8
9 feature_values = (
10     rf_fitted
11     .named_steps["model"]
12     .feature_importances_
13 )
14
15 feature_importance = (
16     pd.DataFrame({
17         "Feature": feature_names,
18         "Importance": feature_values
19     })
20     .sort_values("Importance", ascending=False)
21 )
22
23 display(feature_importance.head(15))
24
25 feature_importance.to_csv(
26     table_path / "final_random_forest_feature_importance.csv",
27     index=False
28 )
29
30 top_features = (
31     feature_importance
32     .head(15)
33     .sort_values("Importance")
34 )
35
36 plt.figure(figsize=(9, 6))
37 plt.barh(
38     top_features["Feature"],
39     top_features["Importance"]
40 )
41
42 plt.title("Top Random Forest Feature Importances")
43 plt.xlabel("Importance")
44 plt.tight_layout()
45
46 plt.savefig(
47     fig_path / "random_forest_feature_importance.png",
48     dpi=300,
49     bbox_inches="tight"
50 )
51
52 plt.show()
```

	Feature	Importance
3	numeric__total_assessments	0.310713
4	numeric__mean_weight	0.259790
2	numeric__mean_score	0.198455
1	numeric__studied_credits	0.019459
10	categorical__code_module_FFF	0.013846
8	categorical__code_module_DDD	0.008271
33	categorical__highest_education_Lower Than A Level	0.008043
15	categorical__code_presentation_2014J	0.007925
0	numeric__num_of_prev_attempts	0.007779
12	categorical__code_presentation_2013B	0.007646
7	categorical__code_module_CCC	0.007066
9	categorical__code_module_EEE	0.006984
6	categorical__code_module_BBB	0.006602
11	categorical__code_module_GGG	0.006097
13	categorical__code_presentation_2013J	0.005563





[21] ✓ 46s

```python
permutation = permutation_importance(
    rf_fitted,
    X_test,
    y_test,
    scoring="f1",
    n_repeats=10,
    random_state=42,
    n_jobs=-1
)

permutation_results = (
    pd.DataFrame({
        "Feature": X_test.columns,
        "Mean_importance": permutation.importances_mean,
        "SD_importance": permutation.importances_std
    })
    .sort_values("Mean_importance", ascending=False)
)

display(permutation_results)

permutation_results.to_csv(
    table_path / "final_permutation_importance.csv",
    index=False
)
```

```
/usr/local/lib/python3.12/dist-packages/joblib/externals/loky/process_executor.py:782: UserWarning: A worker stopped while some jobs were g
  warnings.warn(
```

	Feature	Mean_importance	SD_importance
3	total_assessments	0.226191	0.005213
4	mean_weight	0.154125	0.004652
2	mean_score	0.041162	0.001840
5	code_module	0.032044	0.002560
6	code_presentation	0.001996	0.000815
7	gender	0.001933	0.000531
8	region	0.001799	0.000566
11	age_band	0.001642	0.000736
1	studied_credits	0.001578	0.000524
10	imd_band	0.001524	0.000720
9	highest_education	0.000897	0.000809
0	num_of_prev_attempts	-0.000156	0.000584
12	disability	-0.000378	0.000421

Next steps: New interactive sheet

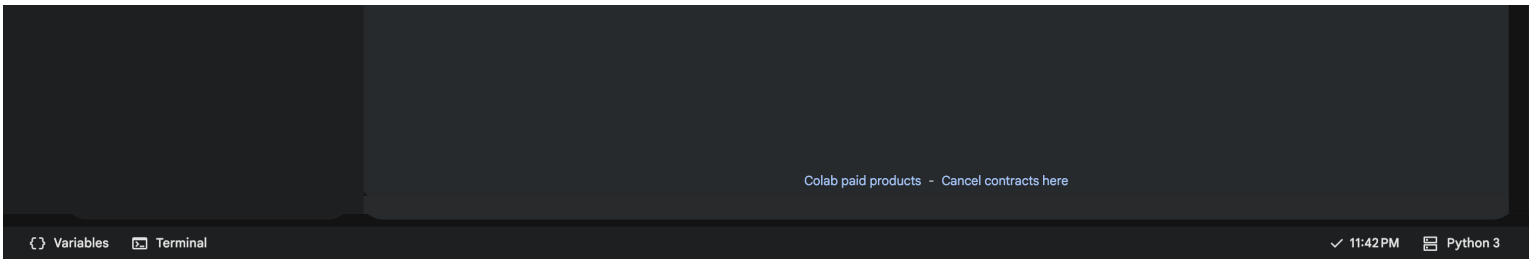
[22] ✓ 0s

```python
final_model_comparison = (
    test_results
    .merge(
        cv_results,
        on="Model",
        how="left"
    )
    .sort_values(
        ["F1", "Recall"],
        ascending=False
    )
)

display(final_model_comparison)

final_model_comparison.to_csv(
    table_path / "final_model_comparison.csv",
    index=False
)

print("Tables saved to:", table_path)
print("Figures saved to:", fig_path)
```

	Model	Accuracy	Precision	Recall	F1	ROC_AUC	Train_time_seconds	Prediction_time_seconds	Mean_Accuracy	SD_Accuracy	Mean_Precis
2	Random Forest	0.9356	0.9690	0.9070	0.9370	0.9786	36.5098	0.2436	0.9365	0.0042	0.9
1	Decision Tree	0.9288	0.9797	0.8835	0.9291	0.9754	0.5638	0.1081	0.9318	0.0038	0.9
0	Logistic Regression	0.9061	0.9405	0.8777	0.9080	0.9514	0.4593	0.0936	0.9093	0.0012	0.9

```
Tables saved to: /content/reports/tables
Figures saved to: /content/reports/figures
```

Next steps: New interactive sheet


Colab paid products - Cancel contracts here
Variables
Terminal
11:42 PM
Python 3